Dhanush S J

dhanushsj2004@gmail.com | Portfolio: Dhanushsj | GitHub: Dhanushsj | LinkedIn: dhanushsj

Summary

Software Engineering student with hands-on experience in Python, SQL, and AI-powered applications, and a strong interest in data intelligence, NLP, and business analytics. Skilled in collecting, cleaning, and transforming structured and unstructured data into AI-ready datasets, with project experience in document processing, multimodal AI, REST APIs, and backend systems. Familiar with data preprocessing, validation, and information retrieval, including LLM-based workflows using Gemini and practical exposure to MongoDB, Firebase, and JSON-based data pipelines. Detail-oriented and quick to learn new technologies, with a passion for building reliable data solutions that support AI systems and business decision-making.

Education

Vellore Institute of Technology — **Expected 2027**
M.Tech Integrated Software Engineering — CGPA: 8.13

NKM Higher Secondary School — **2022**
Class XII — 77%

Desia Matric High School — **2020**
Class X — 93.8%

Technical Skills

Programming: Python, Java, SQL
Data & Analysis: Pandas, Excel, Data Cleaning, Data Preprocessing, Data Validation, Data Structuring, JSON
Web Scraping & Data Collection: BeautifulSoup, Selenium (Basic), REST APIs
Databases: MySQL, MongoDB, Firebase
Knowledge Systems: Knowledge Bases, Vector Databases (Basic), Knowledge Graphs (Basic), Information Retrieval
Backend & Tools: Flask, Node.js (Express), Postman, Git, GitHub
Machine Learning: Scikit-Learn, TensorFlow, PyTorch (Basic)
Cloud: Firebase, GCP (Vertex AI)

Projects

AI Resume Analyzer and ATS Job Matcher | *Python, Flask, Gemini 2.5 Flash, PyPDF2, Tailwind CSS*
- Extracted text from PDF resumes, cleaned noisy content, and converted unstructured documents into labeled candidate profiles for downstream AI analysis.
- Organized parsed resume content into labeled profiles, enabling ATS scoring, resume-to-job matching, and skill-gap analysis.
- Built a Flask web application supporting document uploads, text extraction, metadata generation, and AI-driven resume evaluation.
- Optimized LLM generation parameters to improve consistency, relevance, and accuracy of generated responses.

Real-Time UPI Fraud Detection and Payment Containment System | *Node.js, Express, MongoDB, Razorpay*
- Built a pre-settlement risk scoring engine using velocity detection, sender/receiver trust profiling, and blacklist validation on transaction data.
- Converted high-volume transaction records into risk classifications that guided rule-based payment decisions.
- Implemented a soft-hold mechanism for medium-risk transactions with user confirmation and timeout-based reversal.
- Created an analytics dashboard to visualize fraud trends and transaction patterns, with searchable audit records stored in MongoDB for traceability.

AI-Based Image Processing and Cloud Storage Platform | *Python, Flask, Gemini 1.5 Flash, Cloudinary, Firebase*
- Integrated Gemini 1.5 Flash Vision API to convert uploaded images into descriptive captions and metadata for retrieval and downstream AI workflows.
- Established REST API endpoints for image upload, Cloudinary transformation, and Firebase-based metadata management.
- Implemented user authentication and role-based access to ensure reliable image upload and retrieval operations.
- Optimized API response time by performing Cloudinary transformations asynchronously before database updates.

Event-Driven IoT Surveillance and AI Threat Analysis Pipeline | *Python, ESP32-CAM, Gemini AI, Firebase, Telegram*
- Designed a motion-triggered data collection pipeline that activated ESP32-CAM only during detection events, minimizing redundant data capture.
- Generated natural-language threat summaries with metadata suitable for search, retrieval, and AI-assisted analysis.
- Configured real-time Telegram alerts and centralized Firebase logging for anomaly detection events.
- Structured a modular backend separating device communication, AI inference, and cloud data management for improved maintainability.

AI Voice Assistant with Multilingual NLG/NLU Support | *Python, Gemini 1.5 Flash, SpeechRecognition, pyttsx3*
- Normalized multilingual speech into text and intent representations for reliable downstream AI processing.
- Combined Google Speech-to-Text and pyttsx3 Text-to-Speech with ambient noise adjustment and turn-detection logic for consistent intent recognition.
- Mapped extracted intents to backend response handlers, using Gemini 1.5 Flash for open-ended conversational queries.
- Validated multilingual responses for quality and consistency, refining workflows to improve overall reliability.

Certifications & Achievements

- **Generative AI Essentials: Using LLMs to Work with Data** — *IBM SkillsBuild*; covering LLM-based data analysis, structuring, and information extraction workflows.
- **AI Fundamentals: Foundations for Understanding AI** — *IBM SkillsBuild*; covering AI fundamentals, machine learning concepts, and responsible AI practices.